Skyrim Game Mods Endorsement Prediction with Machine Learning

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***Abstract*—Mods within video games have become an important part of the gaming experience for many, with the activity spawning its own dedicated community of developers and players. We decided to explore one of our questions. What makes a mod popular to others? For our research, we decided to analyze the attributes of the mods on the major PC game title The Elder Scrolls V: Skyrim Special edition. Using natural language processing techniques, we attempted to find out which features affected the popularity of the mod and we did so through the measure of endorsement counts that a mod had. During early research, we found a correlation between the inclusion of adult content within mods and the effect it had on the final endorsement count of the mod. We were able to find out that adult content was indeed an important factor in determining the endorsement count of a mod and hope to answer more questions about the popularity of mods through the analysis of demos and pictures. We use ensemble machine learning model in this study. Our experiment shows that the model's accuracy in predicting the endorsement count of a mod was also very high, yielding a score of 93%. We hope that our research is able to assist the future work of mod and game developers alike.**

***Index Terms*—game prediction, XGBoost, RNN, machine learning**

I. Introduction

First we must introduce the problem that we are trying to solve. Our project attempts to solve a problem in a video game known as The Elder Scrolls V: Skyrim Special Edition (SSE). Within SSE, there are optional community-created modifications to the game known as mods. The word "mod" comes from the word modification [1]. These modifications change the experience of playing the game at many various levels. They are typically created by players of these games using external software tools and are then distributed to other players. These mods are published on a website where they can then be downloaded by other players and be used within the game. Nexus Mods is one of the largest and most popular modding website. In this website, mods are rated by points known as endorsements. Our project attempts to predict the characteristics of a mod that allow it to obtain a high amount of endorsements. For the purposes of this paper, a mod is considered to have a high amount of endorsements if it has more than 100 endorsements. This study aims to test the hypothesis that the inclusion of adult content will allow it to gain more endorsements. We aim to contribute to this field by providing insightful data to future mod and game developers about the features of a mod and its endorsements.

II. Related Work

In this section, we discuss related research work in fields relating to video game mods and the use of neural networks for NLP (Natural Language Processing).

In certain video game communities such as that of Skyrim, mods have become a large part of the experience of playing the game, they add a much needed replay value to the games people love [2]. There is a large motivation from mod creators to create mods for their game with each having their own different reasons. Most of the reasons involved improving certain aspects of the game that they felt could be better. Overall, mods have a large positive impact on gameplay. [3]. Mods have also been identified as possible learning tools as they teach various programming skills to students allowing students to learn concepts as they enjoy playing as well [4].

Natural language processing has many different uses in the field of video games. It could be used as part of the gameplay to analyze speech-based commands that

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the player uses to interact with the world around them [5]. They could also be used in systems used to deduce the sentiment of reviews for games in order to improve gameplay [6]. Natural language processing has also been used in studies attempting to build recommendation systems based on video games. It could be used to recommend games based on the opinions of other gamers [7]. It could be used to recommend obscure games similar to ones a player liked [8].

Bidirectional Encoder Representations from Transformers (BERT) is a transformed based natural language processing technique [9]. All of the neural networks within the architecture used for our research makes use of BERT's state of the art performance in natural language based tasks. BERT makes use of two steps within its framework, pre-training and fine-tuning. It is pre-trained using unlabeled data and then fine-tuned using labeled data from downstream tasks. XGBoost [10] is a revolutionary decision-tree based framework that we used to build our decision forest model. XGBoost's most powerful use is its scalability on every scenario. It allows XGBoost to run multitudes faster on a single machine whilst being able to scale extensively as well.

Recurrent Neural Networks have become popular for tasks such as Natural Language Processing. Recurrent Neural Networks are a type of artificial neural network where connections between units form a directed cycle. This creates an internal state of the network which allows it to exhibit dynamic behavior. Different from other neural networks, RNNs can use their internal memory to process and work on arbitrary sequences of inputs [11]. They have been proven to work well in single task supervised objectives and multitask learning frame works as well [12]. The characteristics of RNNs make it suitable for tasks such as sequential data due to its characteristics of having a "memory" that helps it make its predictions.

Other methods of deducing the reasons behind the popularity of certain mods were explored in other research. It was found that mods with tags seemed to be more popular with the most popular tags being ones focused on visual effects and performance enhancements within the game [13].

The objective of our research differs from those of other research regarding mods such as those finding the motivation of mods [3] or deducing the popularity of mods using other means [13]. Our research objective serves to find to find out the different factors that make up the reasons why a mod may have a higher endorsement count based on its attributes such as adult content. Our research is similar in nature to recommender system classification commonly used to recommend media such as movies. Our research would help those looking to find factors to increasing endorsement count for their mods through patterns in attributes such as a mod's name and its description.

III. Methodology

A. Dataset

Nexus Mods houses a wide variety of mods for various games. There are over 40,900 mods currently available for SSE alone. Therefore, we used this as the primary source for our dataset. We also believe this amount was sufficient enough for us to analyze and create a predictive model from.

To collect and build our dataset, we used the API endpoint provided by Nexus Mods to get the data of the mods[1]. The attributes returned by the response body can be seen in Table I. When analyzing the dataset's distribution of endorsements, we can see that the number of endorsements on mods varies greatly as most mods have less than 200 endorsements with very few outliers containing more or much more. This forces us to deal with the issue with the popularity of mods using endorsements. There may be many patterns that could determine the final endorsement count of a mod. We also needed to determine how many endorsement counts a mod would be required to have before it could be considered to have a lot of endorsements. Additionally, as seen in Figure 1, we were able to find a trend correlating the use of adult content in mods with the average endorsement count of the mod. In certain categories such as category 62, mods with adult content enabled had far more endorsements than ones that did not.

B. Data preparation and processing

Before feeding our dataset to the predictor model for training, we need to handle NaN values as well as removing unnecessary attributes. This was done using Pandas library in Python. The list of attributes we decided to use for training are in Table II. Additionally, we filtered out mods whose *allow_rating* and *status* attributes were *false* and not *published* respectively. Mods not older than six months were also excluded. This is because they were not be a good representation of the dataset and may skew the result of our predictive model.

[1]https://app.swaggerhub.com/apis-docs/NexusMods/nexus-mods_public_api_params_in_form_data/1.0#/Mods/get_v1_games_game_domain_name_mods_id.json

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TABLE I
ATTRIBUTES RETURNED BY THE API ENDPOINT AND THEIR DESCRIPTIONS

Attribute	**Description**
name	Name of the mod.
summary	A short summary of the mod given by the author. Usually 3-4 sentences long.
description	A full description of the mod. It explains the mod in full detail. Can contain a wide variety of information, such as images, videos, tables, and many more.
picture_url	URL of the image used for the mod preview.
uid	The mod's UID stored in the database.
game_id	ID of the game associated with the mod. For SSE, the value is 1704.
allow_rating	Boolean value, whether a mod can be endorsed.
domain_name	Name of the game associated with the game. In the case of SSE, it would be 'skyrimspecialedition'.
category_id	The ID of the category that describes what kind of mod it is (i.e. armor, weapons, new world). The ID for a mod with 'Miscellaneous' category would be 28, for example.
version	the latest version of the mod.
endorsement_count	The amount of endorsements the mod currently has.
created_timestamp	Timestamp of when the mod page was created.
created_time	Time when the mod page was created.
updated_timestamp	Timestamp of when the mod was last updated.
updated_time	Time of when the mod was last updated.
author	The name (or username) of the person that initially created the mod.
uploaded_by	The username of the person that uploaded the mod.
uploaded_users_profile_url	URL to the user profile of the person that uplaoded the mod.
contains_adult_content	Boolean value, determines whether a mod contains adult content.
status	The current status of the mod. There are 6 possible values • published • not_published • wastebinned • removed • hidden • under_moderation
available	Boolean value, determines whether a mod is available for public viewing.
users	Further information regarding the mod uploader.
endorsement	Further information regarding endorsement.

To adjust our dataset for a binary classification problem, we added a new attribute to the dataset called *high*. This attribute describes if a mod is considered to have a high or low endorsement count, with the possible value either 1 or 0 respectively. To determine the borderline for this classification, we took the median of the *endorsement_count* attribute after the previous data cleaning process. The value came up to be 79, and we rounded it off to 100.

The next step we took is to identify and remove outliers. We determined that mods with more than 1000 endorsements are considered to be outliers due to their small count. From this selection we further check the distribution of mods from each category. From all the categories, the median number was 256 and we rounded it off to 300. From here, we only accept categories with more than 300 mods and deem the others as outliers. The distribution of mods for each category can be seen in Figure 2. At the end of the cleaning and filtering, we have over 9,000 rows of data to work with.

C. Model and Techniques

In order to realize our predictive model, we have decided to build our architecture based on an ensemble architecture. Because certain architectures specializes in different formats of data, we decided to create four models tasked on analyzing different features from our dataset. The output from these four models was then fed into one last model tasked on determining the final output. Three of these models were Recurrent Neural Networks (RNNs) tasked on analyzing *name*, *summary*, and *description* features individually. One decision forest model (X1) was used to analyze the remaining features in tabular data format. The output model was a decision

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forest model (X2) to learn from the outputs of the previous models. All of these models were trained for binary classification problem as to allow our predictive model to predict whether a mod for SSE is going to have a high or low endorsement count.

The decision forest models were created using XGBoost, while the RNN models were made using the Keras API that came with the Tensorflow library. All three of the RNN models have the same architecture that utilizes a pre-trained BERT model.The RNN architecture can be viewed on Table III. The text inputs are fed into the BERT encoder, where the words are vectorized based on the dictionary provided by BERT. The models were connected into an ensembled network as shown on Figure 4.

Due to this ensembled architecture, we had to split the dataset accordingly. The *name*, *summary*, and *description* features are separated to their respective RNN models. Everything other feature was fed into the X1 model. To handle class imbalances, we made use of oversampling technique to randomly select which of our data samples to be duplicated, as to not reduce the amount of training samples.

D. Evaluation Method

For the training of the model, first we split the data set into train and test portions. We use an 80:20 ratio for the train test split for every model. For the RNN models, we further split the training dataset with the same ratio for training validation. These models are then trained on one of our desktop computers yielding the following specs.

- GPU: Nvidia RTX 2070
- Processor: Intel i7-9700K
- RAM: 32GB

On this machine, the models are trained for a different number of epochs where a single epoch is a single complete pass through the dataset.

For the evaluation of our predictive model we are using several metrics. We look at the performance of our model using precision, recall, loss and accuracy through a classification report. The graphs we used for further evaluation are the confusion matrix and ROC Curve.

IV. Results and Discussion

Next, we may discuss the results of our predictive model based on our previously mentioned evaluation methods. Beginning with the classification report as shown in Table IV, we may analyze the final model's precision, recall, F1-score, accuracy and support. Within the classification report, there are two classes being analyzed for each metric. The attribute *high* has two values, 0 and 1 where 0 represents a lower endorsement count and 1 represents a higher endorsement count. The classification report shows that for both classes, the X2 model is able to achieve high scores in each metric. For class 0.0, it received a precision score of 91 percent, a recall of 94 percent and a 93 percent f1-score. For class 1.0, it received a precision score of 94 percent, a slightly lower recall with 91 percent and an f1-score of 93 percent. The amount of support for both classes is balanced as well. The model's accuracy is also very high yielding a score of 93 percent. The resulting ROC Curve as shown in Figure 3 shows the True Positive rate against the False Positive rate with the area of the curve being 93 percent. Additionally, after feeding the data to the XGBoost frameworks, we visualized the variable importance within each model to observe the variables with the most effect. Table VII shows the variable importance within model X1 where *contains_adult_content* is the most important variable which matches the trends within the data. Following it are the *summary*, *name* and *description* attributes after that is *category_id*, which matches previous research that creating mods with certain tags may make it more popular [13] or in this case, increase its endorsement count.

TABLE II
Attributes to be used for prediction model training and how to treat their NaN values

Attribute	**NaN Handling Operation and Conditions**
name	Discard row.
summary	Discard row.
description	Discard row.
category_id	Set value as 28 (Miscellaneous).
author	Convert to empty string.
uploaded_by	Convert to empty string.
contains_adult_content	Flag as false.
high	Discard row.

TABLE III
RNN Architecture

	Layers
1	BERT Preprocess
2	BERT Encoder
3	Dropout 0.1
4	Dense 128, Activation Relu
5	Dense 1, Activation Sigmoid

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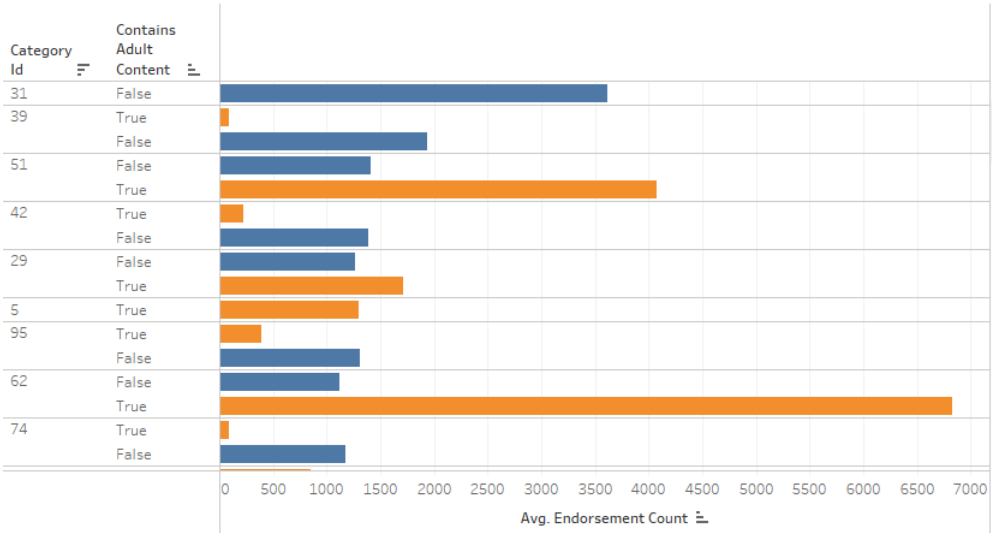

Fig. 1. Average amount of endorsements based on presence of adult content in several mod categories (snippet)

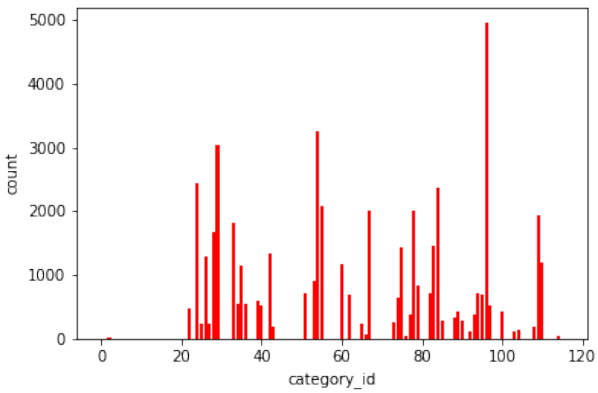

Fig. 2. Category Graph

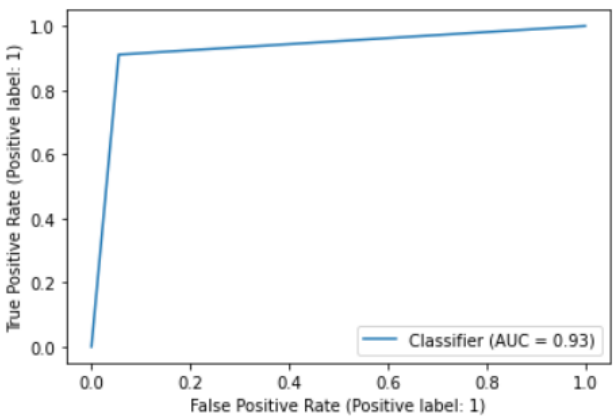

Fig. 3. ROC Curve of X2 Model

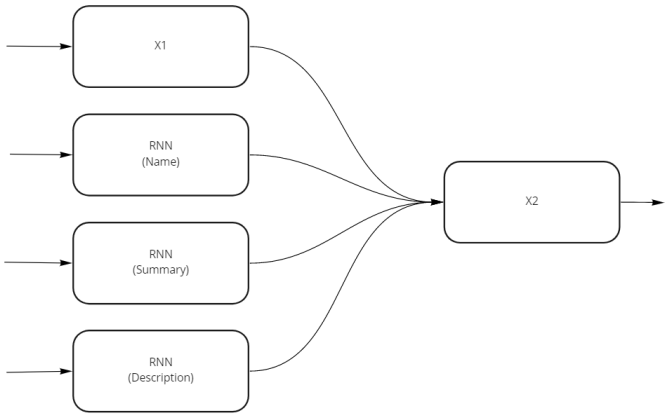

Fig. 4. Ensembled architecture diagram

TABLE IV
CLASSIFICATION REPORT

	Precision	Recall	F1-Score	Support
Low (Class 0)	0.91	0.94	0.93	1143
High (Class 1)	0.94	0.91	0.93	1144
Accuracy			**0.93**	2287
Macro Average	0.93	0.93	0.93	2287
Weighted Average	0.93	0.93	0.93	2287

TABLE V
MODEL ACCURACY SUMMARY

Model	Accuracy
X1	0.80
RNN (Name)	0.60
RNN (Summary)	0.64
RNN (Description)	0.65
X2	0.92

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TABLE VI
VARIABLE IMPORTANCE (X1)

Variable	Importance
contains_adult_content	0.859501
category_id	0.060696
uploaded_by	0.044162
author	0.035641

TABLE VII
VARIABLE IMPORTANCE (X2)

Variable	Importance
X1	0.617784
RNN (Summary)	0.133879
RNN (Name)	0.131725
RNN (Description)	0.116612

TABLE VIII
VARIABLE IMPORTANCE (OVERALL)

Variable	Importance
contains_adult_content	0.530986
summary	0.133879
name	0.131725
description	0.116612
category_id	0.037497
uploaded_by	0.027283
author	0.022018

V. CONCLUSION AND RECOMMENDATION

In conclusion, we believe our initial hypothesis is proved by the results of our predictive model. It can be deduced that mods may receive a higher endorsement count through adult content. First we analyzed the data received from the Nexus Mods website. We found that there was a largely uneven distribution for the endorsement counts of mods and the correlation between the average endorsement count of a mod with the enabling of adult content. After trimming down the data and feeding it through our designed architecture, we found that generally many features affected the final endorsement count of a mod with adult content being a large factor as well.

In the future, we plan to consider the other factors that might attract users on the Nexus Mods website to endorse a mod, such as demos and pictures, in our study. We hope that this study in video game mods will lead to better insight from mod and video game developers in creating the best possible experiences for the games and media that they create.

SUPPLEMENTARY SOURCE CODE

To see the source code head to the following link: https://gitlab.com/giantsweetroll/mod-success-predictor

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